STA_445_Assignment 7

Kellen Phelps

Load your packages here:

Problem 1:

The `infmort` data set from the package `faraway` gives the infant mortality rate for a variety of countries. The information is relatively out of date, but will be fun to graph. Visualize the data using by creating scatter plots of mortality vs income while faceting using `region` and setting color by `oil` export status. Utilize a $\log_{10}$ transformation for both `mortality` and `income` axes. This can be done either by doing the transformation inside the `aes()` command or by utilizing the `scale_x_log10()` or `scale_y_log10()` layers. The critical difference is if the scales are on the original vs log transformed scale. Experiment with both and see which you prefer.

a. The `rownames()` of the table gives the country names and you should create a new column that contains the country names. *`rownames`

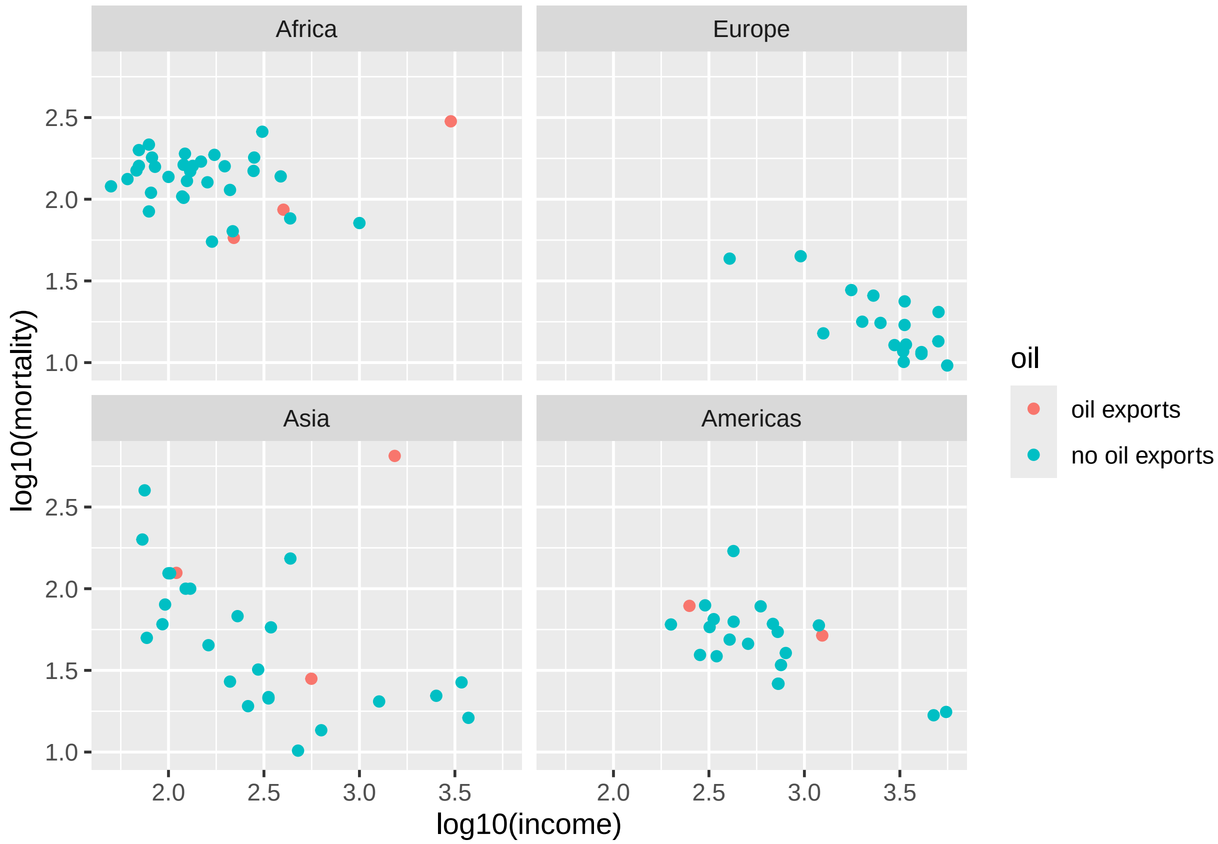
```
infmort2 <- infmort %>% mutate(country = rownames(infmort) %>% trimws())
head(infmort2)
```

```
##                        region income mortality            oil   country
## Australia                Asia   3426      26.7 no oil exports Australia
## Austria                Europe   3350      23.7 no oil exports   Austria
## Belgium                Europe   3346      17.0 no oil exports   Belgium
## Canada               Americas   4751      16.8 no oil exports    Canada
## Denmark                Europe   5029      13.5 no oil exports   Denmark
## Finland                Europe   3312      10.1 no oil exports   Finland
```

b. Create scatter plots with the `log10()` transformation inside the `aes()`command.

```
ggplot(infmort2, aes(x=log10(income),y=log10(mortality),color=oil)) +
  geom_point() +
  facet_wrap(vars(region))
```

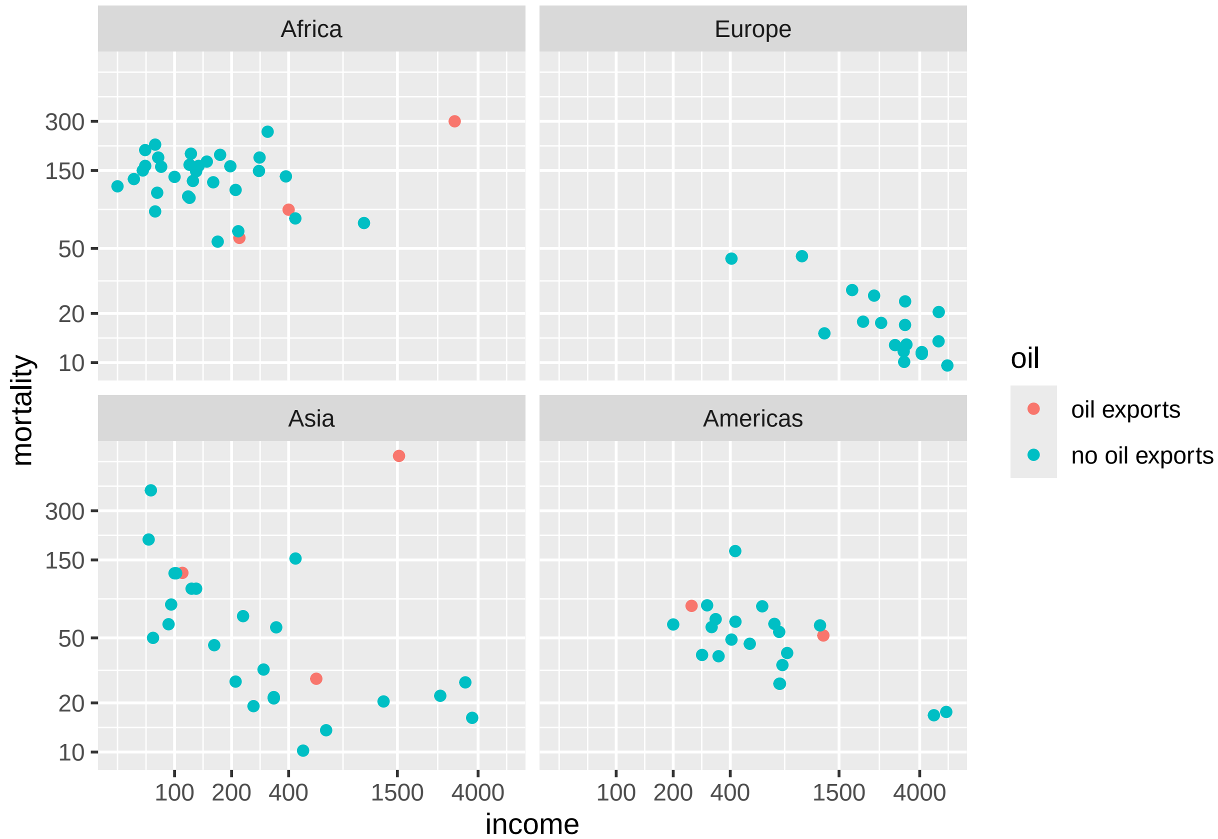
```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```



c. Create the scatter plots using the `scale_x_log10()` and `scale_y_log10()`.Set the major and minor breaks to be useful and aesthetically pleasing.Comment on which version you find easier to read.

```
ggplot(infmort2, aes(x=income,y=mortality,color=oil)) +
  geom_point() +
  facet_wrap(vars(region)) +
  scale_x_log10(breaks=c(100,200,400,1500,4000)) +
  scale_y_log10(breaks=c(10,20,50,150,300))
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```

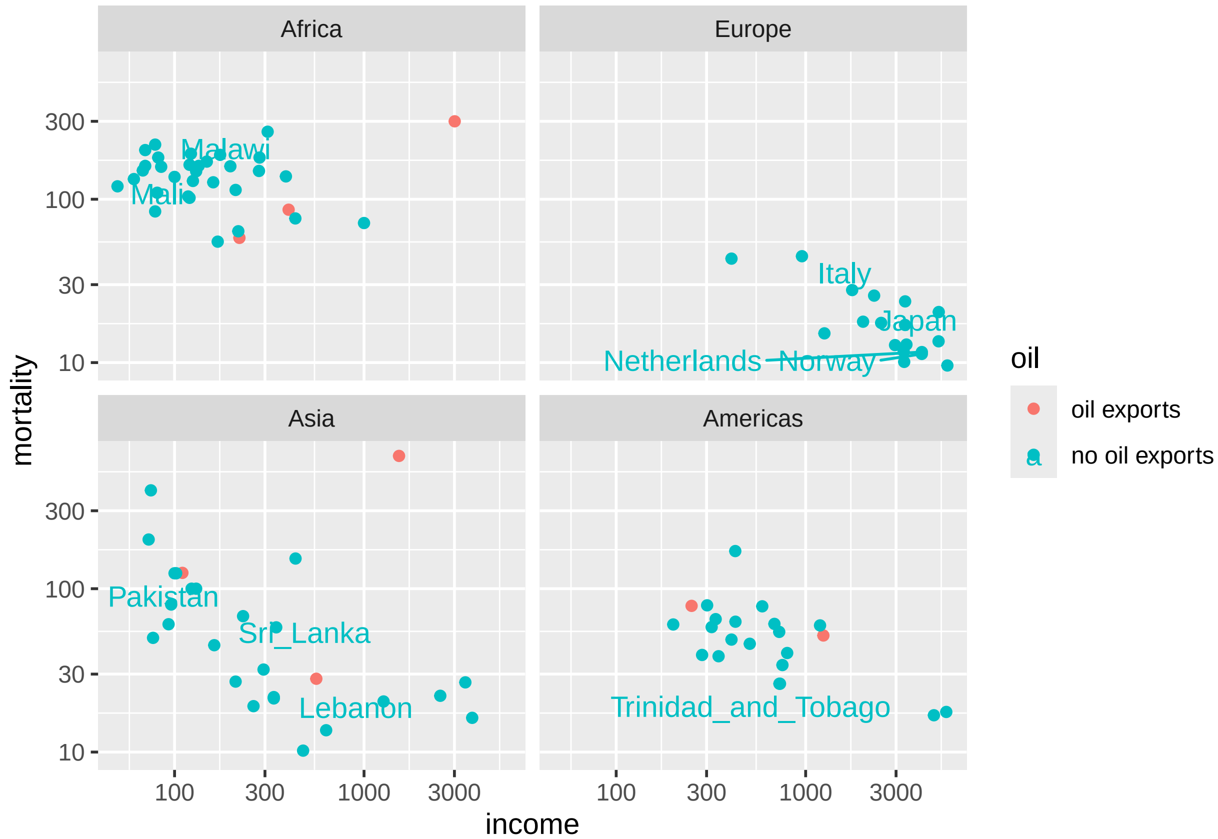


The second graph is easier to read as it has axes in terms of the original variable instead of the logged variable.

d. The package `ggrepel` contains functions `geom_text_repel()` and `geom_label_repel()` that mimic the basic `geom_text()` and `geom_label()`functions in `ggplot2`, but work to make sure the labels don't overlap. Select 10-15 countries to label and do so using the `geom_text_repel()` function.

```
selected_countries <- slice_sample(infmort2,n=10)
ggplot(infmort2, aes(x=income,y=mortality,color=oil,label=country)) +
  geom_point() +
  facet_wrap(vars(region)) +
  scale_x_log10() +
  scale_y_log10() +
  geom_text_repel(data=selected_countries,aes(x=income,y=mortality))
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Problem 2

Using the `datasets::trees` data, complete the following:

a. Create a regression model for $y =$ `Volume` as a function of $x =$ `Height`.

```
model <- lm(Volume ~ Height,trees)
model
```

```
##
## Call:
## lm(formula = Volume ~ Height, data = trees)
##
## Coefficients:
## (Intercept)       Height
##     -87.124        1.543
```

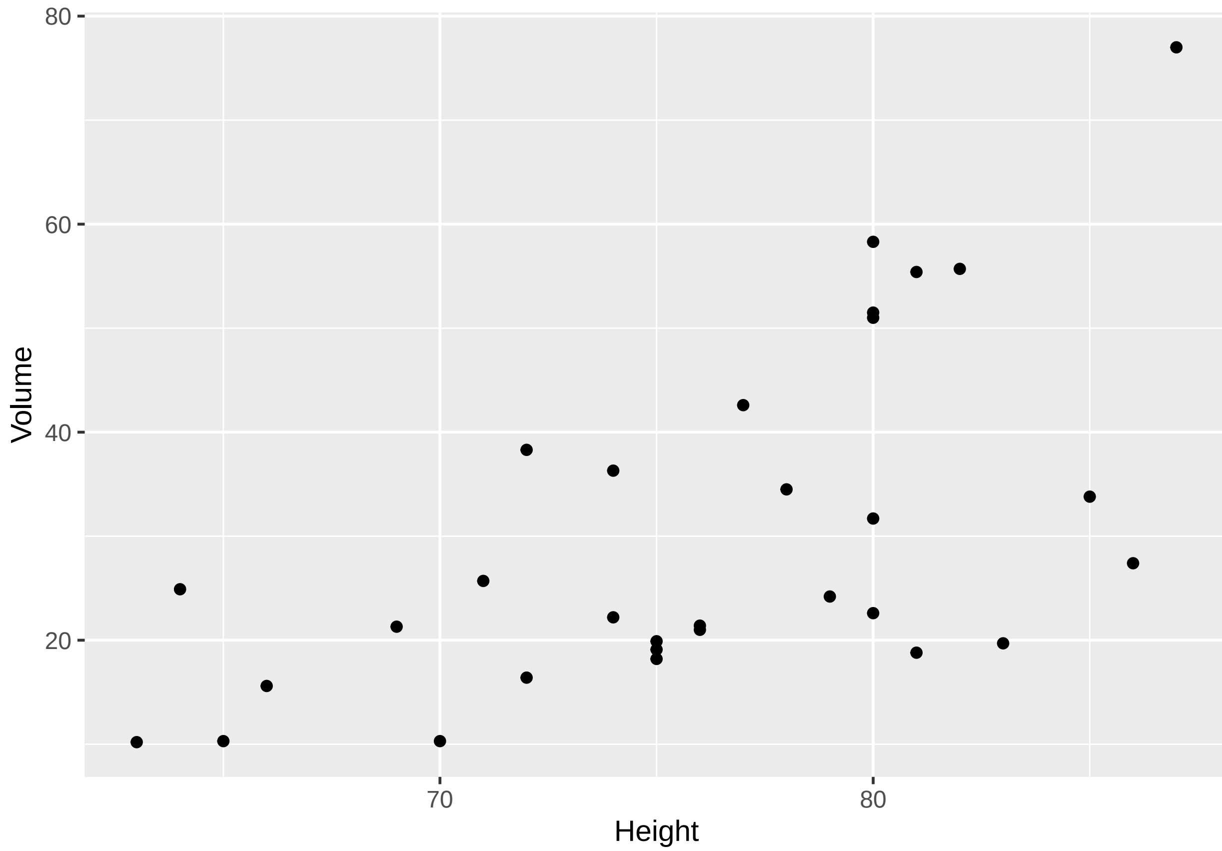
b. Using the str(your model's name) command, to get a list of all the information stored in the linear model object. Use $ to extract the slope and intercept of the regression line (the coefficients).

```
model$coefficients
```

```
## (Intercept)      Height
##   -87.12361     1.54335
```

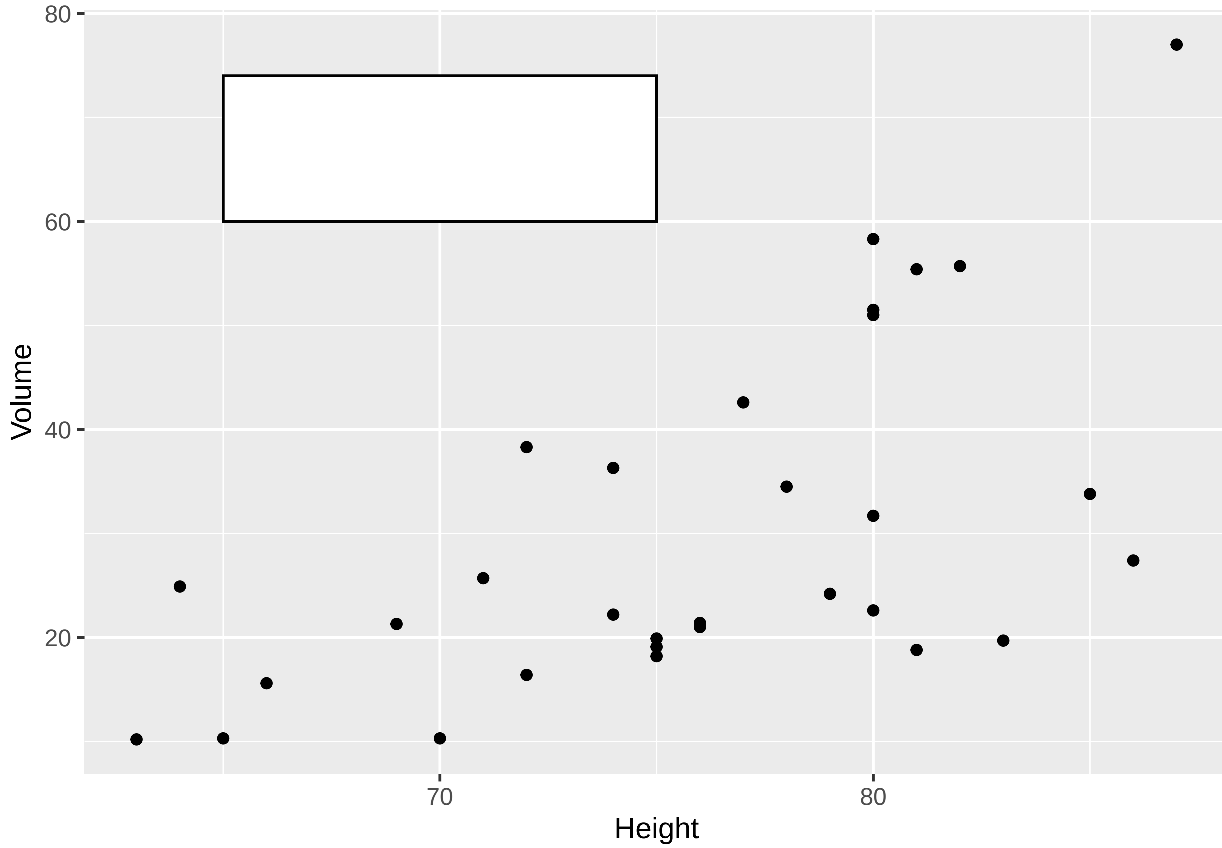
c. Using `ggplot2`, create a scatter plot of Volume vs Height.

```
p1 <- ggplot(trees, aes(x=Height, y=Volume)) +
  geom_point()
p1
```



d. Create a nice white filled rectangle to add text information to using by adding the following annotation layer.

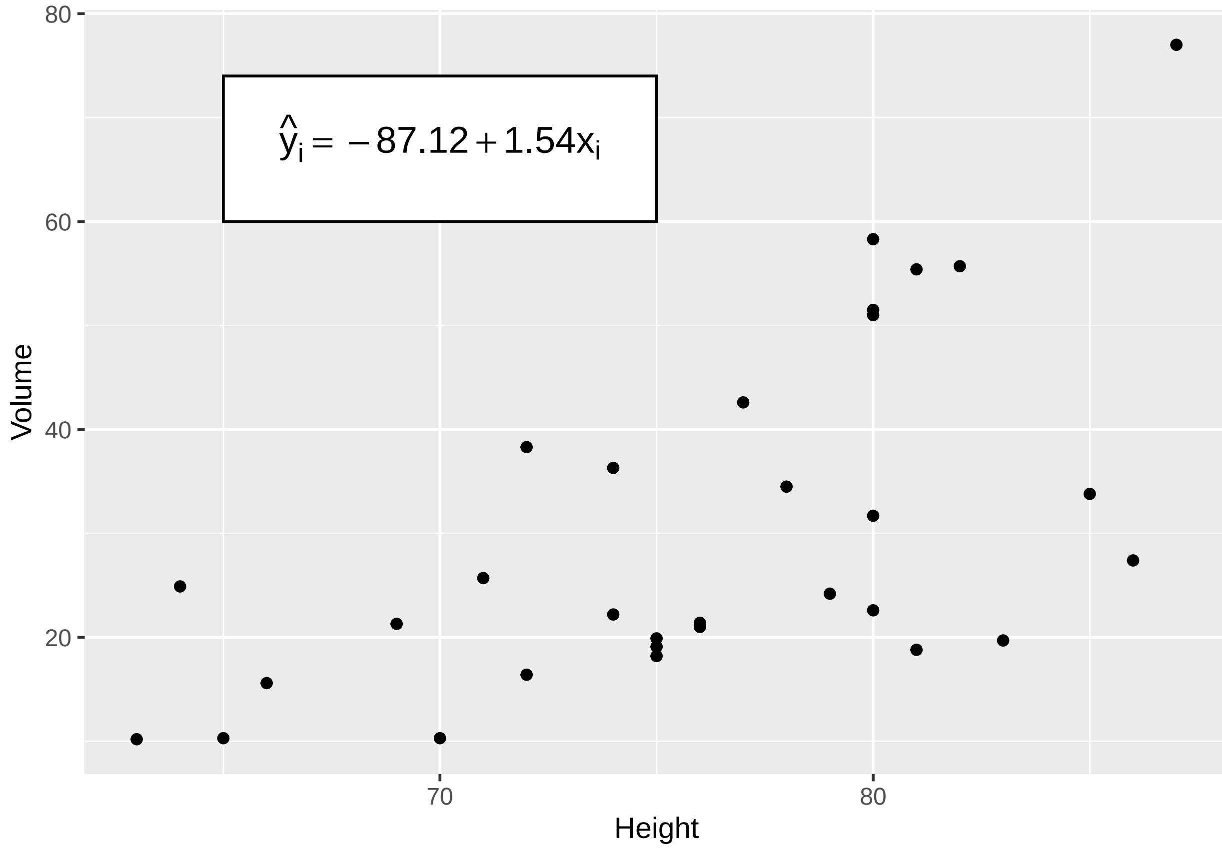
```
p2 <- p1 +
  annotate('rect', xmin=65, xmax=75,
           ymin=60, ymax=74,
           fill='white', color='black')
p2
```



annotate('rect', xmin=65, xmax=75, ymin=60, ymax=74, fill='white', color='black')

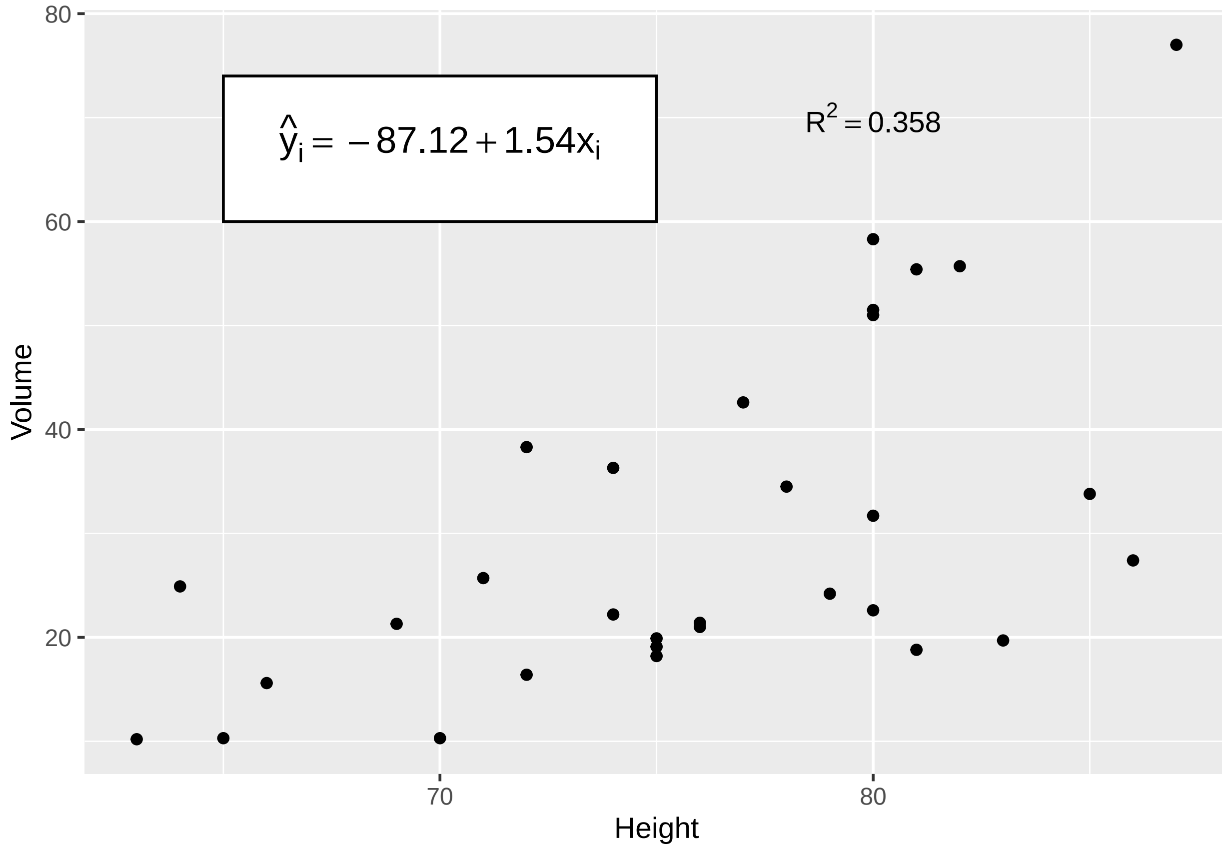
e. Add some annotation text to write the equation of the line $\hat{y}_i = -87.12 + 1.54x_i$ in the text area.

```
p3 <- p2 + annotate('text',
                    label=TeX("$\\hat{y}_i = -87.12 + 1.54x_i$",
                              output="character"),
                    parse=TRUE,
                    x=70,y=68,
                    color='black',size=5)
p3
```



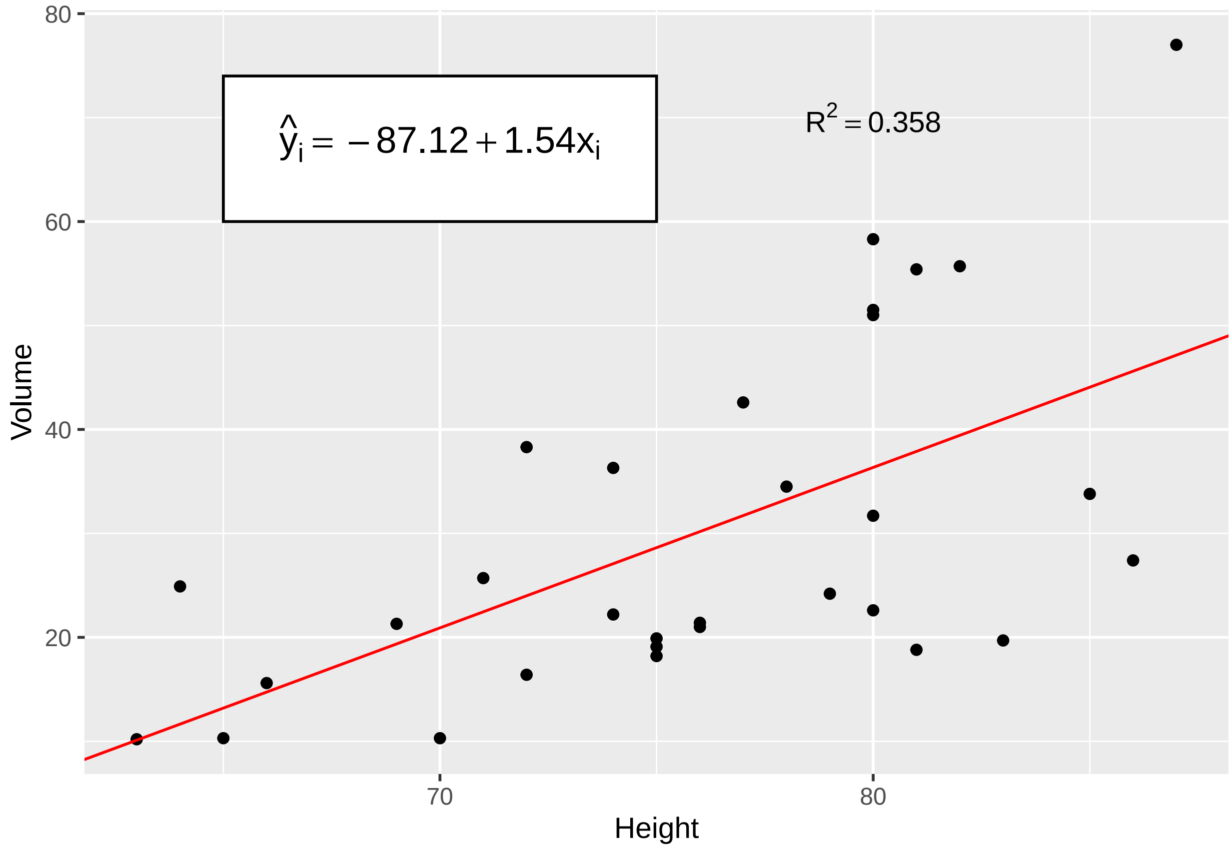
f. Add annotation to add $R^2 = 0.358$

```
p4 <- p3 + annotate('text',
                    label=TeX("$R^2 = 0.358$",output="character"),
                    parse=TRUE,x=80,y=70)
p4
```



g. Add the regression line in red. The most convenient layer function to use is `geom_abline()`.

```
p5 <- p4 + geom_abline(aes(
  intercept=model$coefficients[1],
  slope=model$coefficients[2]),
  color='red')
p5
```



Problem 3

In `datasets::Titanic` table summarizes the survival of passengers aboard the ocean liner *Titanic*. It includes information about passenger class, sex, and age (adult or child). Create a bar graph showing the number of individuals that survived based on the passenger `Class`, `Sex`, and `Age` variable information. You'll need to use faceting and/or color to get all four variables on the same graph. Make sure that differences in survival among different classes of children are perceivable. *Unfortunately, the data is stored as a `table`and to expand it to a data frame, the following code can be used.*

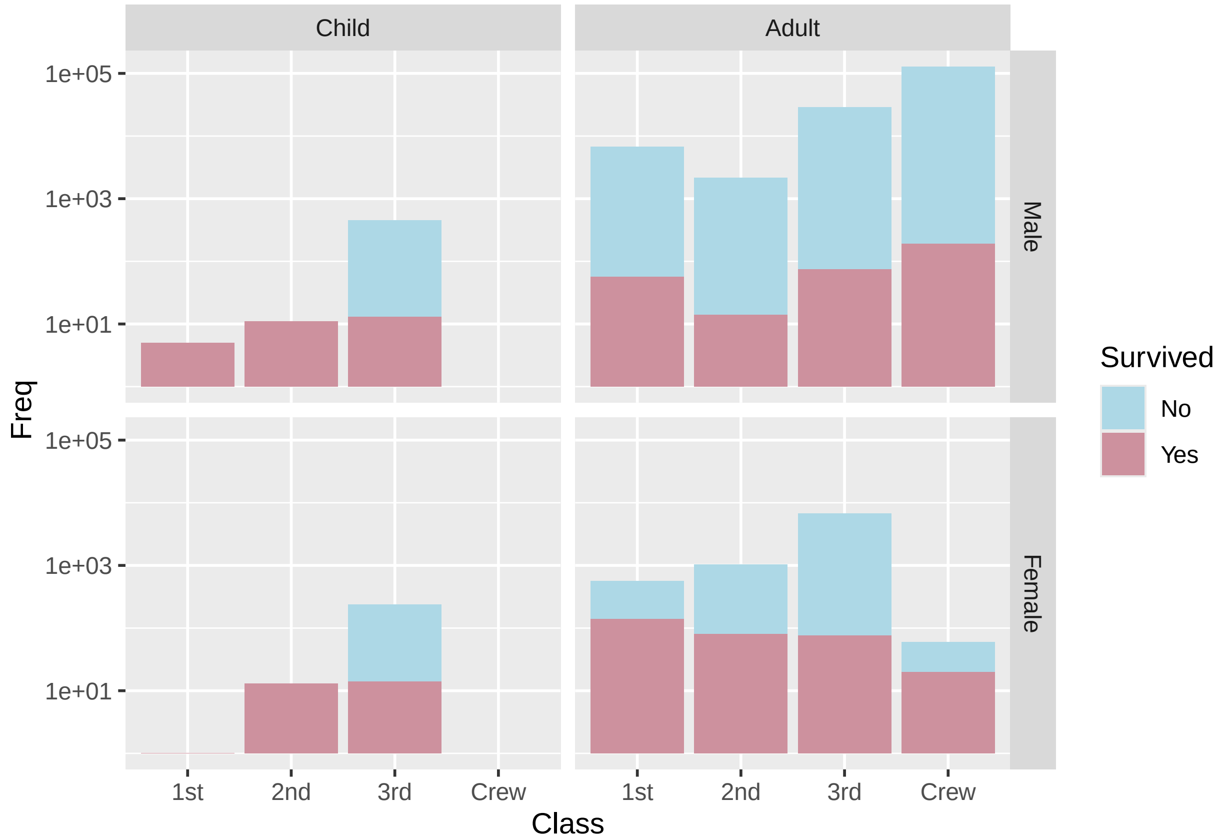
```
‘‘‘ r
Titanic <- Titanic %>% as.data.frame()
‘‘‘
```

a. Make this graph using the default theme. *If you use color to denote survivorship, modify the color scheme so that a cold color denotes death.*

```r
titplot <- ggplot(data=Titanic,aes(x=Class)) +
  facet_grid(rows=vars(Sex),cols=vars(Age)) +
  geom_bar(stat="identity",aes(y=Freq,fill=Survived)) +
  scale_fill_manual(values=c("lightblue","pink3")) +
  scale_y_log10()
titplot
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```

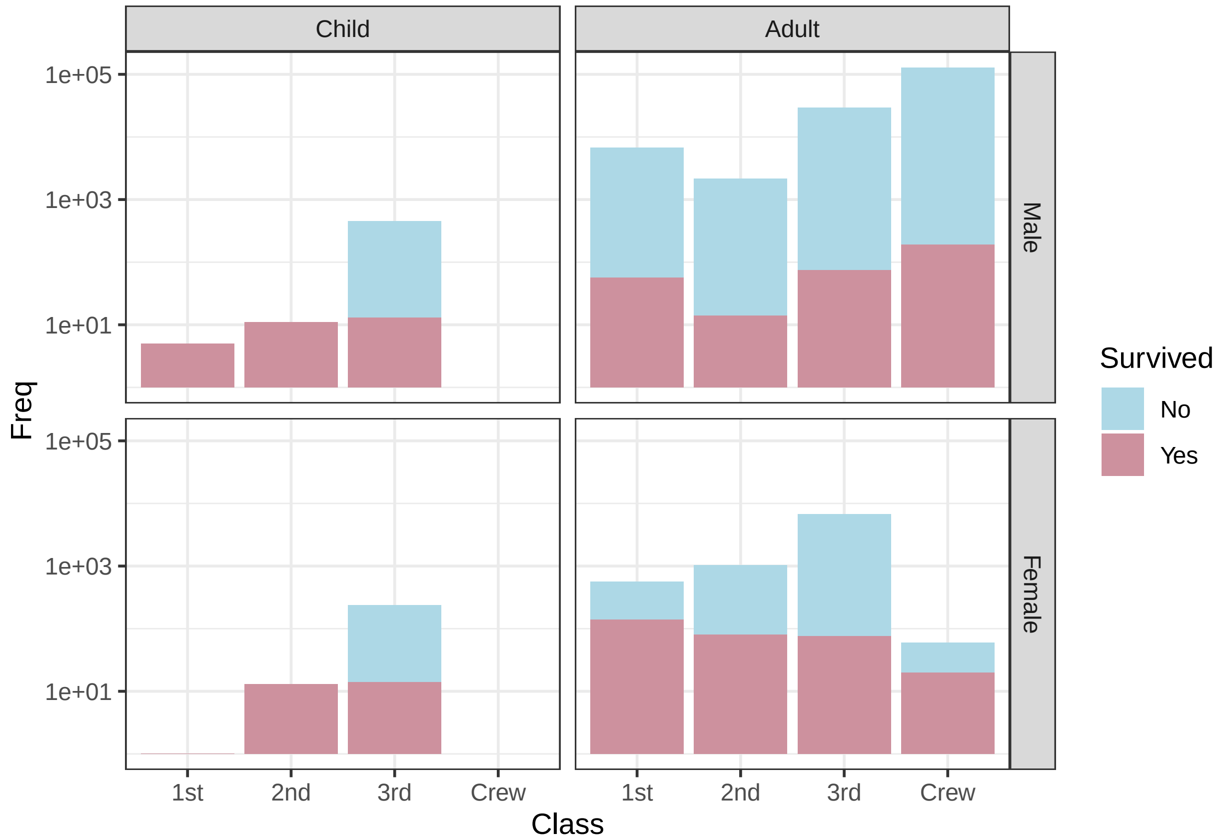


b. Make this graph using the `theme_bw()` theme.

```r
titplot + theme_bw()
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```

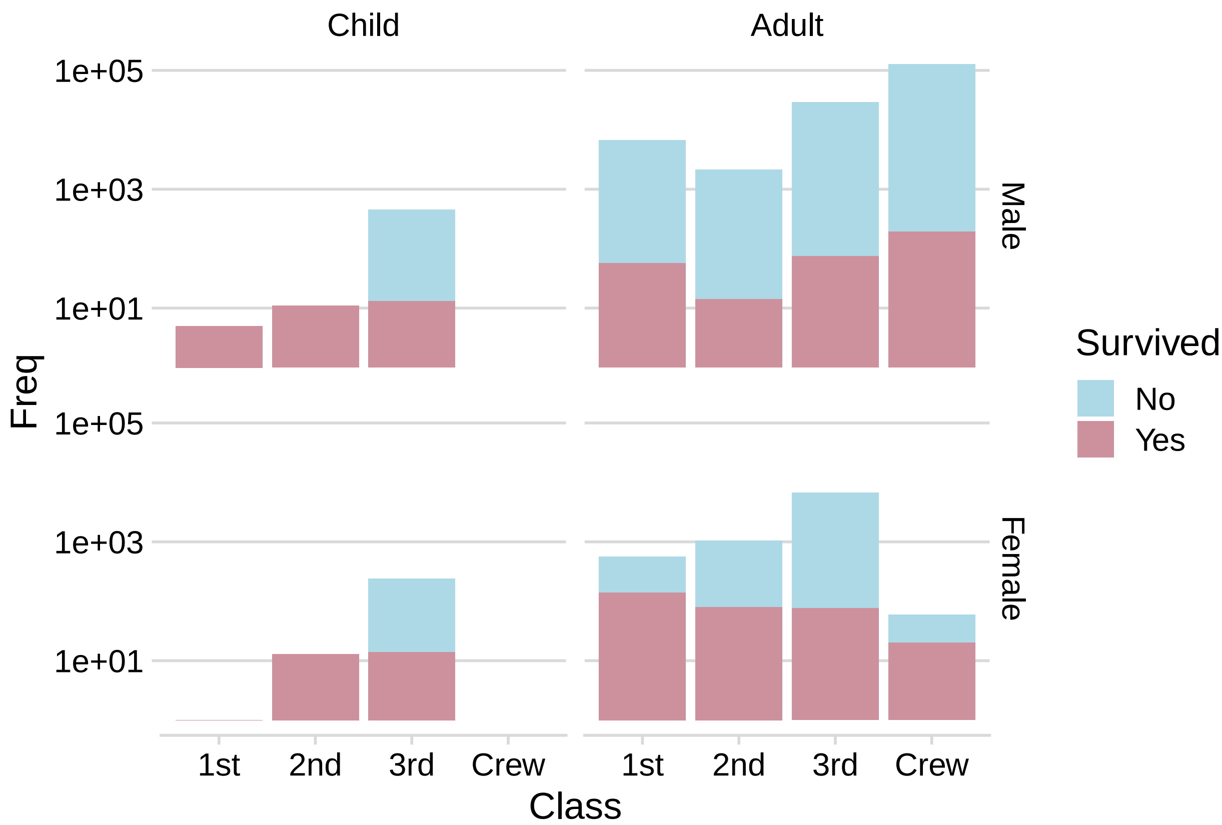


c. Make this graph using the `cowplot::theme_minimal_hgrid()` theme.

```
titplot + theme_minimal_hgrid()
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_bar()`).
```



d. Why would it be beneficial to drop the vertical grid lines?

The x-axis here is categorical rather than quantitative, and which bar goes to which class is very obvious without the lines. Getting rid of them removes visual clutter and improves readability.